Confidence Intervals: Introduction

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Introduction

A point estimate without an interval is an answer without error bars – nearly useless for decision-making. A **confidence interval** is a range of values, computed from data, that would contain the true parameter in a specified fraction of hypothetical repeated samples. This "coverage" interpretation is precise and useful but different from what most readers casually assume.

Prerequisites

The reader should understand the concept of a sampling distribution and of a standard error.

Theory

A $1-\alpha$ confidence interval for a parameter θ is a pair of statistics (L, U) depending on the data such that

$$P\{L \leq \theta \leq U\} \geq 1-\alpha \quad \text{for every } \theta,$$

where the probability is over repeated sampling. The conventional choice is $\alpha = 0.05$, giving 95% CIs.

Interpretation. A 95% CI is a realisation of a random procedure that captures the true parameter 95% of the time across hypothetical repetitions. A specific computed interval either contains θ or does not; the "95%" refers to the procedure, not to the single interval. This is often misstated as "there is a 95% probability that θ lies in the interval" – correct only in a Bayesian sense, where the object is a **credible interval** computed from a posterior distribution.

Construction techniques.

1. **Pivotal quantities**: if $Q(X, \theta)$ has a distribution not depending on θ, then $P\{a \leq Q \leq b\} = 1-\alpha$ for known a, b, and inversion gives a CI for θ. The classic example: for iid normal data, $(\bar{X}-\mu)/(s/\sqrt{n}) \sim t_{n-1}$, giving the t-interval.
2. **Normal approximation**: for asymptotically normal estimators, $\hat{\theta} \pm z_{1-\alpha/2}\widehat{\mathrm{SE}}(\hat{\theta})$. Simple and widely used; coverage depends on how close the sampling distribution is to normal.
3. **Likelihood-ratio intervals**: invert the LRT at each candidate value; the set where it is not rejected is a CI. Better coverage than Wald in many cases; symmetric on the log-likelihood scale, not the parameter scale.
4. **Bootstrap intervals**: resample the data, compute the statistic, and use quantiles (percentile method) or bias-corrected variants for the interval. Works when an analytical SE is unavailable.

Assumptions

Each construction has its own assumptions. The t-interval requires normal data or a large enough sample for CLT. Wald intervals require asymptotic normality of $\hat{\theta}$ and a well-estimated SE. Bootstrap intervals require exchangeable observations and inherit the sampling design.

R Implementation

```
library(boot)

set.seed(2026)
x <- rnorm(40, mean = 75, sd = 12)

t.test(x)$conf.int

n <- length(x)
xbar <- mean(x); s <- sd(x)
xbar + c(-1, 1) * qnorm(0.975) * s / sqrt(n)

boot_mean <- function(d, i) mean(d[i])
b <- boot(x, boot_mean, R = 5000)
boot.ci(b, type = c("perc", "bca"))
```

We compute a t-interval, a Wald (normal-approximation) interval, and two bootstrap intervals for the same sample mean.

Output & Results

```
95 percent confidence interval:
 71.34  78.68     # t-interval

[1] 71.51 78.51   # Wald

Percentile: (71.52, 78.53)
BCa:        (71.59, 78.60)
```

The four intervals agree closely in this well-behaved example; they differ more for heavily skewed statistics or small samples.

Interpretation

For a manuscript: "The mean was 75.0 (95% CI 71.3 to 78.7, t-interval)." The CI is interpreted as the range of parameter values consistent with the data under the t-sampling model; values outside the interval would be rejected by a 5% level two-sided test.

Practical Tips

- Prefer CIs over p-values when reporting an effect; they convey both the effect size and its uncertainty.
- For small samples or skewed distributions, bootstrap BCa intervals usually have better coverage than Wald.
- For proportions near 0 or 1, use Wilson or Clopper-Pearson, not Wald; the latter can extend outside $[0, 1]$.
- The level $1 - \alpha$ is a choice, not a universal truth; pre-specify it and justify it in the methods.
- Do not interpret a specific 95% CI as "95% probability the parameter is inside" unless you are doing Bayesian analysis.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**

- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr